

USRP B210 SDR Link — System Reference

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USRP B210 SDR Link — System Reference

End-to-end digital communication system for two USRP B210 radios (UHD), written in C++. This document lists **every feature, algorithm (with the math and the common alternatives), and command-line option** exposed by `sdr_system`.

- Binary: `build/sdr_system`
- Help: `./sdr_system --help`

- Two operating styles: **one-way** (`--role tx / --role rx`, repeated sends, auto-terminating RX) and **ARQ** (`--role source_arq / --role sink_arq`, stop-and-wait with ACK).

1. Signal-chain overview

TX: message $\rightarrow$ split into chunks $\rightarrow$ CRC-16 $\rightarrow$ [FEC encode] $\rightarrow$ bits $\rightarrow$ symbols (modulator)
 ├ single-carrier: RRC pulse-shape $\rightarrow$ preamble prepend $\rightarrow$ USRP TX
 └ OFDM: QAM $\rightarrow$ IFFT+CP frame [SC-sync | chan-est | data] $\rightarrow$ USRP TX

RX: USRP RX $\rightarrow$ energy detector (burst gate) $\rightarrow$ AGC $\rightarrow$ frame/symbol sync
 ├ single-carrier: matched filter $\rightarrow$ timing (Gardner) $\rightarrow$ [equalizer] $\rightarrow$ demod
 └ OFDM: Schmidl-Cox sync + CFO $\rightarrow$ FFT $\rightarrow$ 1-tap eq $\rightarrow$ pilot CPE $\rightarrow$ demod
 $\rightarrow$ [FEC decode] $\rightarrow$ CRC-16 check $\rightarrow$ (ARQ: ACK if OK) $\rightarrow$ reassemble message

Default physical parameters: symbol rate 0.8 MHz, U/D = 2/1 $\rightarrow$ sample rate 1.6 MHz (integer **2 samples/symbol** on the wire), RRC roll-off 0.25, 151 taps.

2. Modulation

`--scheme <NAME>` selects the constellation. Supported (bits/symbol in parentheses):

Family	Schemes	bits/sym
PSK	BPSK (1), QPSK (2), 8-PSK (3)	1-3
QAM	16-QAM (4), 32-QAM (5), 64-QAM (6), 128-QAM (7), 256-QAM (8)	4-8
Differential	DBPSK (1), DQPSK (2), 8-DPSK (3), PI4-QPSK (2)	1-3
APSK	16APSK (4), 32APSK (5)	4-5

(16QAM and 16-QAM spellings both accepted.)

Math. A modulator maps $k = \log_2 M$ bits to one of M complex points $s = I + jQ$:

- M-PSK**: points on the unit circle, $s = e^{j2\pi m/M}$, $m = 0, \dots, M-1$. All symbols equal-energy; demod by nearest angle. Robust but spectrally inefficient at high order (points crowd on the circle).
- M-QAM**: square/cross grid of amplitudes, $s = (2i - \sqrt{M} + 1) + j(2q - \sqrt{M} + 1)$. Best power efficiency for a given rate on an AWGN channel, but needs accurate amplitude $\rightarrow$ sensitive to gain/EVM/clipping (this is why 16-QAM failed OTA at $\sim 20\%$ EVM while QPSK survived $\sim 37\%$).
- APSK**: points on a few concentric rings. Lower PAPR than QAM, favored on non-linear (satellite) amplifiers — a middle ground between PSK and QAM.
- Differential (D*)**: information is in the **phase change** between consecutive symbols, $s_n = s_{n-1} e^{j\Delta\varphi}$. No absolute phase reference needed $\rightarrow$ tolerates carrier phase offset / no PLL, at a ~ 3 dB SNR penalty. PI4-QPSK rotates the constellation by $\pi/4$ each symbol to avoid zero-crossings (lower envelope variation).

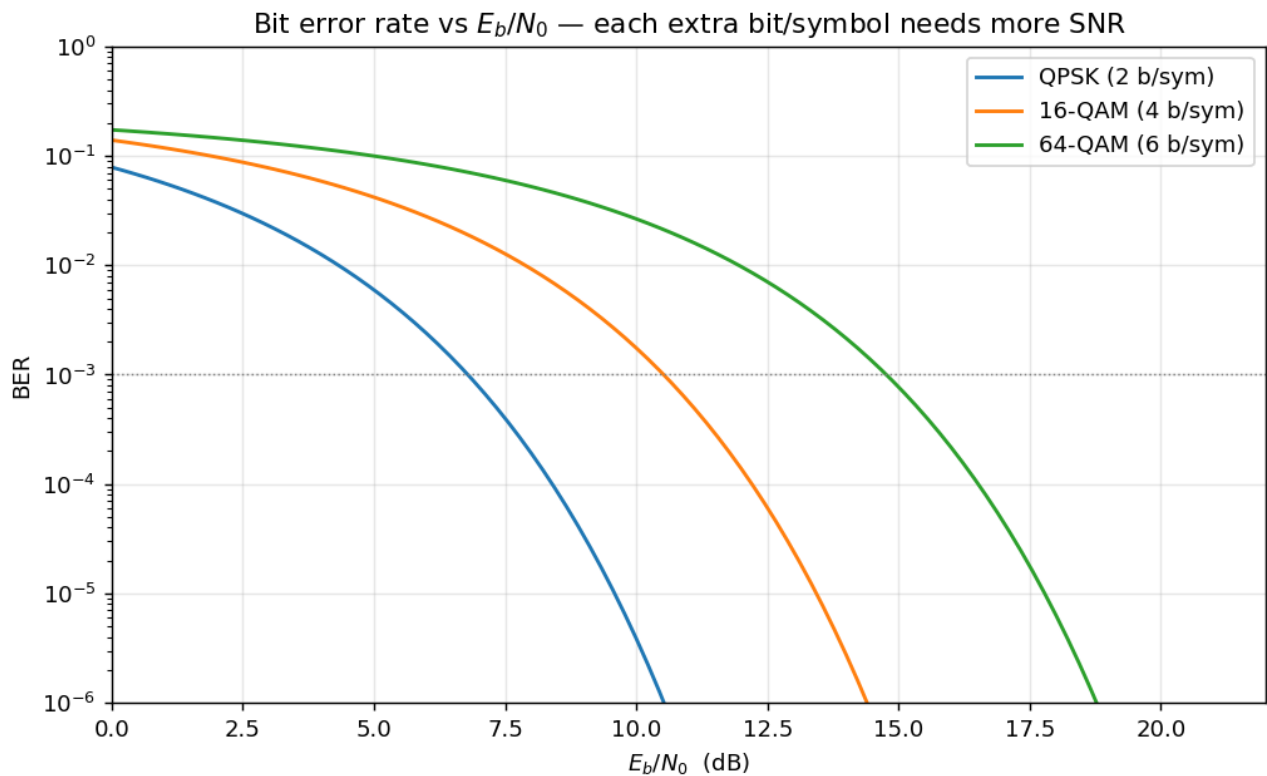
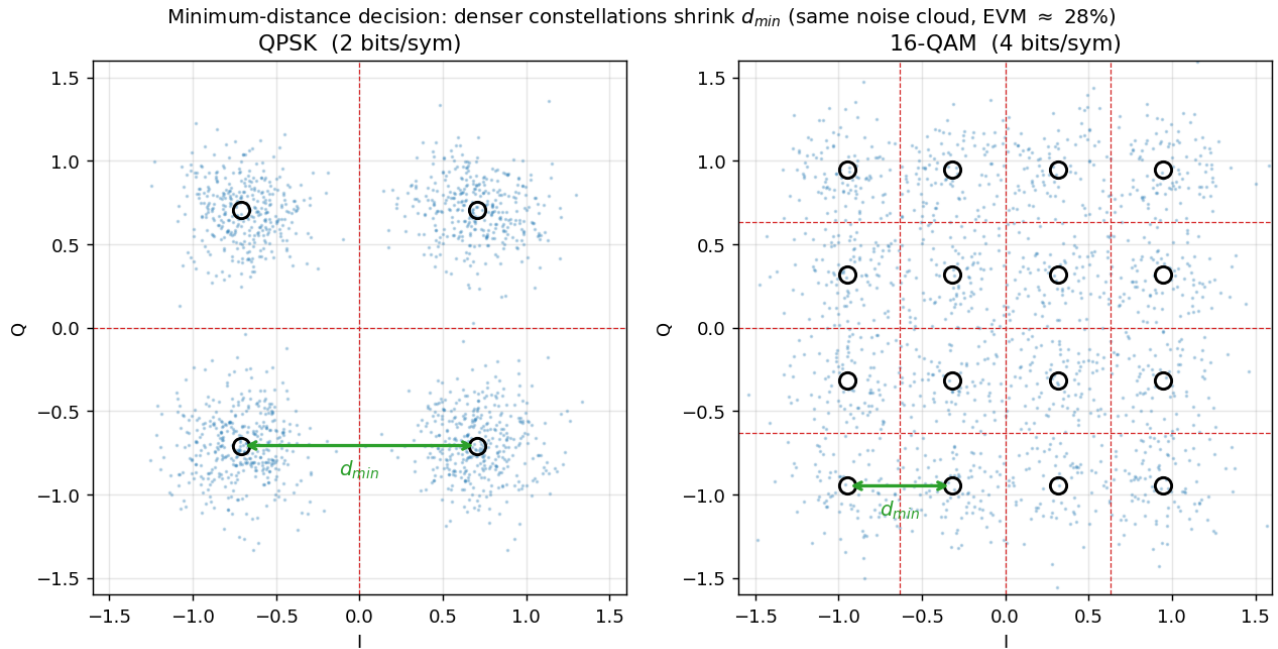
Detection & why constellation order costs SNR. The demodulator makes a **minimum-distance decision** — pick the constellation point nearest the received symbol y (optimal for equiprobable symbols in AWGN):

$$\hat{s} = \arg \min_{c \in \mathcal{C}} |y - c|$$

An error occurs when noise/distortion pushes y past the halfway line to a neighbour, i.e. beyond half the **minimum distance** $d_{\min}/2$. For a fixed average power E_s , packing more points shrinks $d_{\min}$: QPSK has $d_{\min} = \sqrt{2E_s}$ with the whole plane split into 4 quadrants, while 16-QAM squeezes 16 points into the same power so $d_{\min}$ is $\sim 3\times$ smaller and its decision cells $\sim 3\times$ tighter. The **error vector magnitude**

$$\text{EVM} = \frac{\text{rms}(y - \hat{s})}{\text{rms}(c)}$$

measures how far symbols land from ideal; reliable decoding needs roughly $\text{EVM} \lesssim d_{\min}/(2 \text{ rms})$, which is why QPSK survived $\sim 37\%$ EVM over the air but 16-QAM (needing $\lesssim 12\%$) did not. **Gray coding** the bit->point map (adjacent points differ by one bit) makes each symbol error cost only ~ 1 bit error.



Bits $\leftrightarrow$ symbols detail. Gray-like index mapping; `bits_to_index` zero-fills a partial final group so schemes whose bits/symbol don't divide the frame (32-QAM, 128-QAM) still align — a bug fixed early in this project.

Common alternatives not implemented: GMSK/CPM (constant-envelope, used in GSM/Bluetooth), non-square 8-QAM, probabilistic constellation shaping.

3. Waveform: single-carrier vs OFDM

`--waveform sc` (default) or `--waveform ofdm`.

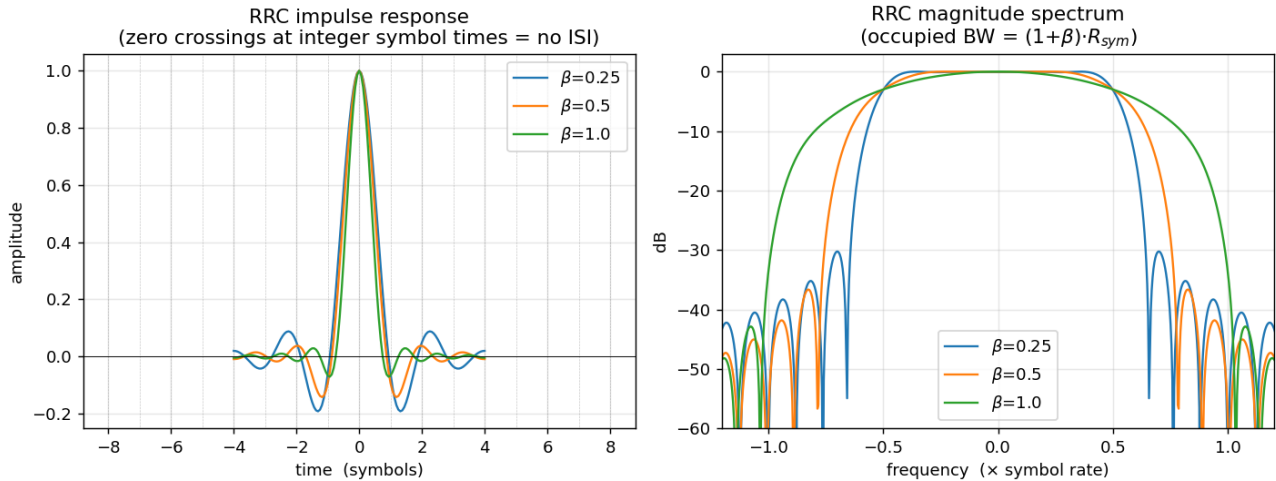
3.1 Single-carrier + RRC pulse shaping

Symbols are upsampled U/D and convolved with a **root-raised-cosine** filter (`--filter_type rrc`, `--roll_off 0.25`, `--num_taps 151`).

Math. A raised-cosine (RC) spectrum satisfies the **Nyquist ISI criterion** — its impulse response is zero at every other symbol instant, $g(mT) = \delta[m]$ — so neighbouring symbols don't interfere at the sampling times. Splitting it as **RRC at TX and RRC at RX** makes the cascade $G_{RRC}(f) G_{RRC}(f) = G_{RC}(f)$ a full RC (matched-filter optimal in AWGN) and maximizes SNR. The roll-off $\beta \in [0, 1]$ trades occupied bandwidth against sidelobe decay:

$$B = (1 + \beta) \frac{R_{sym}}{2} \quad (\text{one-sided baseband; occupied width} = (1 + \beta) R_{sym}).$$

Alternatives: `rc` (raised cosine, all shaping at TX), `lp` (plain low-pass); Gaussian shaping (GMSK) is the common constant-envelope alternative.



3.2 OFDM

Frame layout per burst: [Schmidl-Cox sync symbol | channel-estimation symbol | data symbols...], each symbol = N -point IFFT + cyclic prefix.

- `--ofdm-fft 64` — number of subcarriers N (FFT size).
- `--ofdm-cp 16` — cyclic-prefix length (must exceed the channel's delay spread).
- `--ofdm-tx-peak 0.5` — scales the frame so its peak $\approx$ this value (OFDM has high PAPR; keep the DAC out of clipping).

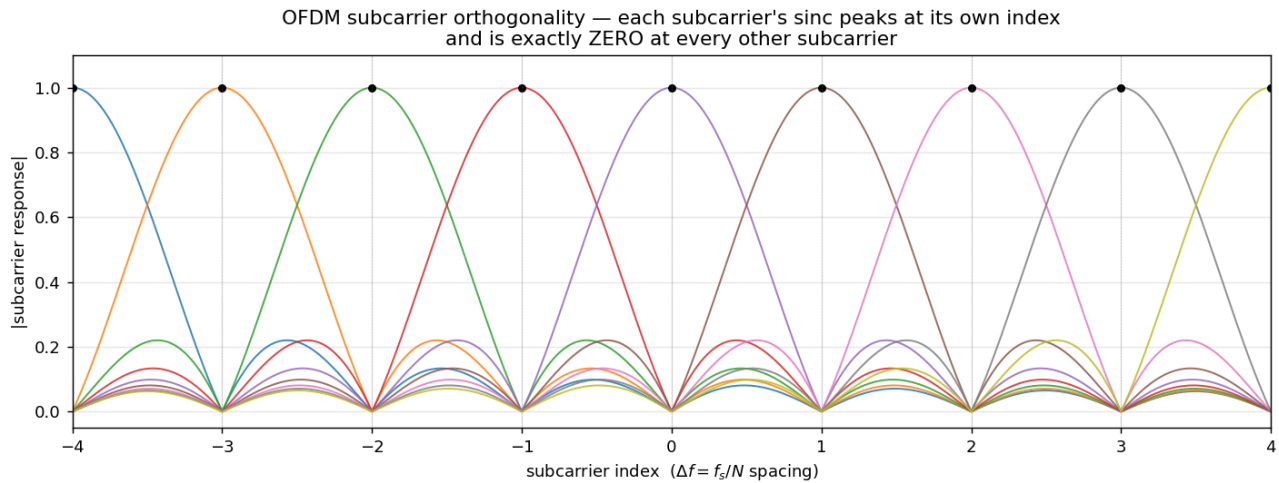
Math. OFDM splits the band into N narrow orthogonal subcarriers via the inverse DFT:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j2\pi kn/N}.$$

A cyclic prefix (copy of the tail prepended) turns the channel's *linear* convolution into *circular* convolution, so after the FFT each subcarrier sees a single complex gain $H[k]$:

$$Y[k] = H[k] X[k] + W[k].$$

Equalization is then **one complex divide per subcarrier**, $\hat{X}[k] = Y[k]/H[k]$ — no FIR equalizer needed even under heavy multipath, provided the delay spread < CP. This is why OFDM is preferred for dense QAM on frequency-selective channels. Cost: high peak-to-average power ratio (PAPR) and sensitivity to carrier frequency offset (loss of subcarrier orthogonality -> inter-carrier interference).



Channel estimation & pilots. One known **channel-estimation symbol** (known value on every active subcarrier) gives $H[k] = Y_{\text{chest}}[k]/\text{ref}[k]$. **Scattered pilots** (every 8th active subcarrier carries a known value) then track the residual **common phase error (CPE)** that grows symbol-by-symbol from residual CFO.

CPE estimation — the fix that made OFDM work OTA. Per data symbol, estimate the phase ϕ from the pilots and derotate:

$$\hat{\phi} = \arg\left(\sum_{k \in \text{pilots}} Y[k] H^*[k] \text{PILOT}^*\right) \quad (\text{maximum-ratio combining})$$

The earlier version used the *equalized* pilot $Y[k]/H[k]$, which **blows up on a deep-fade subcarrier** (tiny $|H[k]|$) — one garbage pilot then hijacked ϕ and smeared the whole symbol into a blob (EVM ~67%, 0 CRC over the air; invisible in a mild simulation channel). Weighting by channel power $|H[k]|^2$ via $Y \cdot \text{conj}(H)$ suppresses faded pilots instead of amplifying them -> four clean clusters (EVM ~37%), message decodes. Disable with env OFDM_NO_CPE=1 to compare.

Common alternatives: decision-feedback / per-subcarrier phase-locked tracking, MMSE channel estimation (vs the least-squares one used here), pilot-aided residual-CFO (SFO) slope correction, windowed-OFDM / filtered-OFDM / OTFS for spectral containment.

4. Energy detection (burst gating)

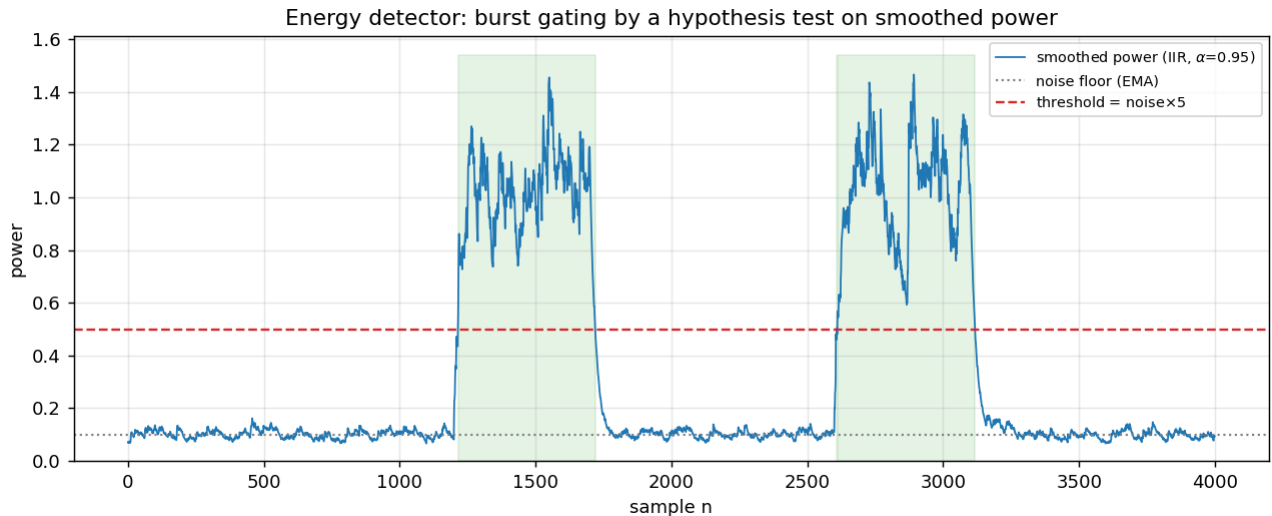
The RX runs continuously; the energy detector decides *when a burst is present* so downstream sync isn't run on noise.

Math. This is a **binary hypothesis test** per sample — H_0 : noise only, H_1 : signal present — on the received power. The instantaneous power $|r[n]|^2$ is smoothed by a one-pole IIR (exponential moving average) to reduce variance:

$$P[n] = (1 - \alpha) |r[n]|^2 + \alpha P[n - 1], \quad \text{declare burst if } P[n] > \gamma.$$

Larger α = more averaging (time constant $\approx 1/(1 - \alpha) \approx 20$ samples), which trades detection latency for a lower false-alarm rate — the smoothing collapses the variance of the noise-power estimate so a single threshold γ cleanly separates the two hypotheses. ($\alpha = 0.02$ barely smoothed $\rightarrow$ the detector fired on every noise spike and even chopped real bursts apart on the RRC envelope; $\alpha = 0.95$ gives one clean capture/burst.)

- **Adaptive threshold** (`--det-adaptive true`, default): $\gamma = \nu \cdot \hat{N}_0$ where $\hat{N}_0$ is the measured noise floor and $\nu = \text{--det-mult } 5$ (raise to 10–30 over the air so ambient RF doesn't trigger it; too high misses weak bursts).
- **Continuous noise tracking** (`--det-continuous true`, default): the noise floor itself is re-estimated by an EMA during idle periods, $\hat{N}_0 \leftarrow (1 - \beta)\hat{N}_0 + \beta |r[n]|^2$, so it follows drifting ambient noise. Alternative: one-shot startup calibration (`--det-continuous false`) — simpler but brittle if the environment changes.
- **Fixed threshold** (`--det-adaptive false + --det-threshold 1e-7`): absolute cutoff, only for a known clean cabled link.
- `--energy_packet_size` — how many samples to grab once a burst is detected (auto-sized per modulation).



Common alternatives: matched-filter / correlation detection (detect on the *known preamble* rather than energy — more sensitive but needs the template), constant-false-alarm-rate (CFAR) detectors, cyclostationary feature detection (works below the noise floor).

5. Synchronization, frequency & phase offset

5.0 The received-signal model (what every block below is undoing)

After the RX front-end down-converts to baseband, the transmitted symbol stream $s[m]$ (pulse-shaped, sampled at f_s) arrives distorted by four separable impairments:

$$r[n] = \underbrace{A e^{j(2\pi \frac{\Delta f}{f_s} n + \varphi_0)}}_{\text{carrier}} \underbrace{\sum_m s[m] g(nT_s - mT - \tau)}_{\text{timing (delay } \tau)} + \underbrace{w[n]}_{\text{noise}}$$

- A — unknown gain -> removed by the **AGC** (§9).
- Δf — **carrier frequency offset (CFO)**: the two radios' oscillators differ (each B210 TCXO is ± 2 ppm, so up to ~ 4 ppm ≈ 3.6 kHz at 915 MHz). A constant Δf produces a phase that **grows linearly with time**, $\theta[n] = 2\pi(\Delta f/f_s) n$ — it *spins* the constellation. Left uncorrected it turns the constellation into a ring.
- φ_0 — **static carrier phase offset**: the oscillators' phase difference at acquisition plus the channel's phase. A constant *rotation* of the whole constellation.
- τ — **timing offset**: the ADC sampling grid isn't aligned to the symbol centers, so the matched filter is sampled off its peak -> inter-symbol interference (ISI).
- $w[n]$ — AWGN.

Note the coupling: **residual CFO becomes a phase ramp** $\varphi[n] = \varphi_0 + 2\pi(\Delta f/f_s) n$, which is why §5.5 uses a *second-order* loop (tracks a constant phase **and** its constant rate) and why OFDM needs per-symbol CPE tracking (§3.2). The recovery order is: **frame sync** -> **timing** -> **CFO** -> **phase**.

5.1 Preamble (the known reference all estimators key off)

--preamble m-sequence (default) or --preamble zadoff; --m 5 sets the m-sequence order (length $L = 2^m - 1 = 31$).

A good preamble p has a **thumbtack autocorrelation** $\sum_n p[n] p^*[n+k] \approx E \delta[k]$ (large at zero lag, near-zero elsewhere) so the correlator (§5.2) gives one sharp, unambiguous peak.

- **m-sequence** — maximal-length binary (BPSK ± 1) shift-register sequence. Periodic autocorrelation is two-valued: L at zero lag, -1 otherwise. Real-valued.
- **Zadoff-Chu** — complex **CAZAC** sequence $p[n] = e^{-j\pi u n(n+1)/L}$: *constant amplitude* and *zero cyclic autocorrelation*. The flat spectrum makes it ideal for training a complex channel estimate / equalizer, and its constant envelope correlates cleanly even under CFO. Preferred for equalizer training and dense QAM.

For **differential** schemes the **last preamble symbol doubles as the differential reference**: the TX encodes the first data symbol relative to it, and the RX keeps that one symbol in front of the data so the differential decoder returns exactly N symbols (not $N - 1$). This keeps the decoded bit count aligned with the fixed FEC/CRC/ARQ framing, and these schemes bypass the coherent phase PLL (§5.5) — a decision-directed loop would pin the absolute phase and destroy the transitions they carry.

5.2 Frame synchronization — ACQ correlation

The receiver must find where the packet starts. `ACQSynchronizer::SamplesACQPerformance` slides the known preamble over the burst and computes, at each candidate offset τ (the `ComputeCorrelation` routine), the **matched-filter** / **cross-correlation** magnitude:

$$R(\tau) = \left| \sum_{n=0}^{L-1} p^*[n] r[\tau + n o_s] \right| \quad (o_s = \text{samples/symbol})$$

By the matched-filter theorem this is the optimal detector for a known sequence in AWGN. At the true start τ^* , all L terms add coherently -> $R(\tau^*) \approx \sum |p[n]|^2$; at any wrong offset the terms add with random phases -> $R \sim \sqrt{L}$ (much smaller). Detection:

- **Coarse pass**: find τ where $R(\tau)$ first crosses --sync_threshold (default 15), then
- **Fine pass**: search $\pm a$ few samples around it for the true maximum.
- **Noise floor / CFAR**: `EstimateNoiseFloor` correlates away from the peak (excluding a guard zone) to measure the off-peak floor, so the threshold can be set relative to it. Set --sync-threshold below the real peak ($\approx$ preamble length ~ 31 after AGC) but above that floor; watch the [ACQ] Peak correlation log lines.

Because the search tests **every sample** (not every os -th), the peak location simultaneously gives **coarse symbol timing** — the correct sub-symbol sampling phase. (Striding by os would test only one of the os phases and could lock a half-symbol off -> catastrophic ISI, BER ≈ 0.5 . This was an actual bug; the brute-force search fixed it.)

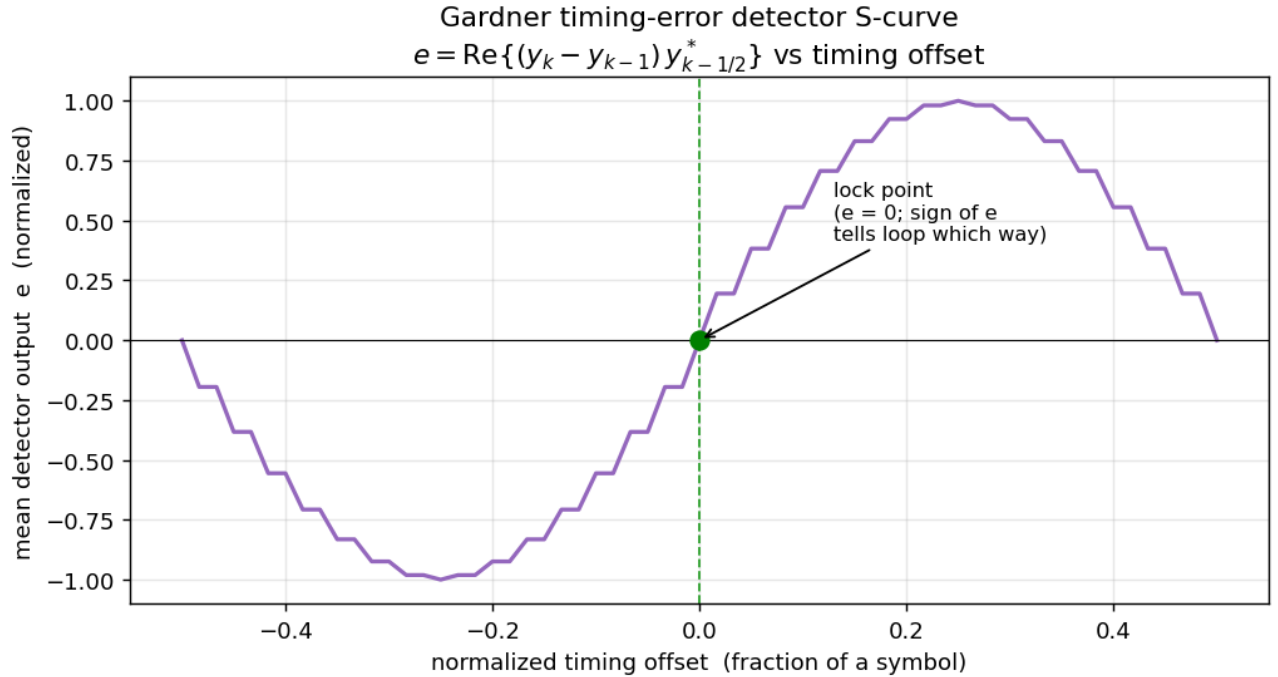
5.3 Symbol-timing recovery — Gardner TED

The ACQ peak gets timing right to the nearest sample; the **Gardner timing-error detector** (GardnerTED) then tracks the residual *fractional* delay continuously. `--timing_loop_bw 0.015` (loop bandwidth $B_n \cdot T$), `--timing_damping 0.707`, `--sps_sync 5`.

Error detector. With two samples per symbol — one at the symbol center y_k and one at the half-symbol midpoint $y_{k-1/2}$ — the Gardner error is

$$e_k = \text{Re}\left\{ (y_k - y_{k-1}) y_{k-1/2}^* \right\}.$$

Intuition: if sampling is early or late, the midpoint sample $y_{k-1/2}$ (sitting on the transition between symbols) correlates with the slope $y_k - y_{k-1}$; at perfect timing the midpoint is a true zero-crossing and $e_k = 0$. It is **decision-independent and carrier-phase-independent** (the $\text{Re}\{\cdot\}$ with the difference term cancels a constant rotation), so it works *before* CFO/phase are resolved — the reason Gardner is the default choice for QAM.



Loop. The NCO advances $\omega = 2/\text{sps}$ per input sample (two strobes/symbol — using $1/\text{sps}$ gives only one strobe and the "midpoint" lands a full symbol away -> the loop tracks garbage; this was a fixed bug). A **Farrow interpolator** produces the fractional-delay sample at offset μ , and a **proportional-integral (PI) loop filter** drives μ :

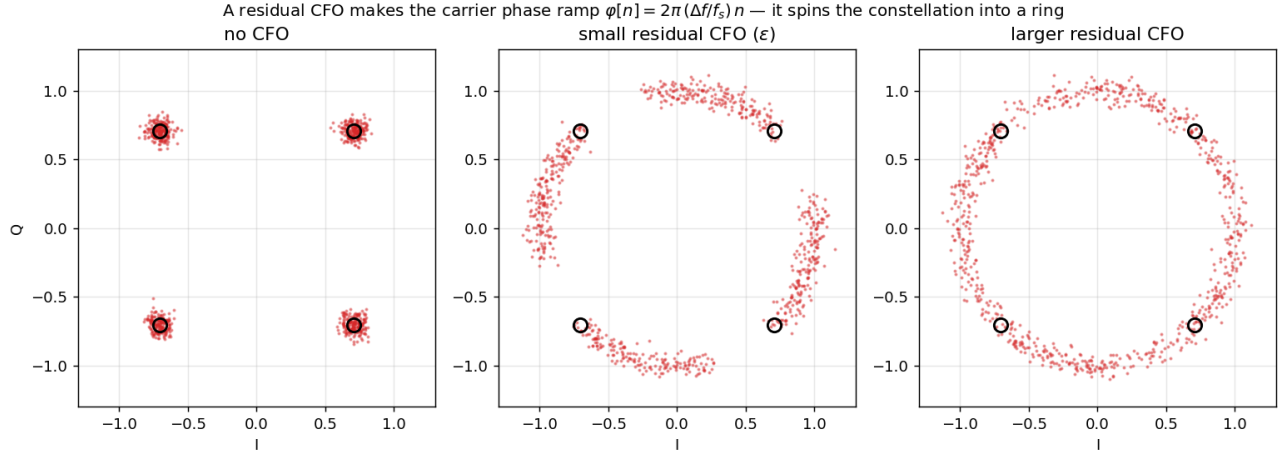
$$\nu \leftarrow \nu + K_2 e_k \quad (\text{integrator — tracks a constant timing rate / clock offset}), \quad \mu \leftarrow \mu + K_1 e_k \quad (\text{proportional}).$$

with ν clamped to $\pm 10\%$ of ω . The PI coefficients come from the standard second-order design (same θ_n/ζ formulas as §5.5). Only the symbol-center strobes are emitted -> exactly one sample per symbol out.

Alternatives: Mueller & Müller (decision-directed, needs only 1 sample/sym but needs carrier first), early-late gate, zero-crossing TED.

5.4 Carrier frequency offset (CFO) estimation & correction

A constant CFO spins the constellation at $2\pi \Delta f / f_s$ rad/sample. Two estimators are provided (CF0Estimator), both **data-aided** off the preamble, then FrequencyShifter removes it by multiplying sample n by $e^{-j2\pi(\Delta f/f_s)n}$ (a running phase accumulator, so blocks join seamlessly).



(A) Pilot-aided (phase-slope across preamble symbols). For preamble symbols $p[k]$ received at $r[k \cdot \text{sps}]$, the phase advance per symbol is measured with the known data stripped off:

$$\varphi_k = \arg(r[k \text{ sps}] r^*[(k-1)\text{sps}] p^*[k] p[k-1]), \quad \widehat{\Delta f} = \frac{f_s}{2\pi \text{ sps}} \overline{\varphi_k}.$$

Averaging over the preamble reduces noise. Unambiguous range $|\Delta f| < f_s/(2 \text{ sps})$ (one symbol of phase must not wrap).

(B) Auto-correlation (Moose / Schmidl-Cox). With a preamble made of two identical halves of length L , the second half equals the first times the CFO phase accrued over L samples:

$$P = \sum_{n=0}^{L-1} r[n] r^*[n+L] \approx |A|^2 L e^{j2\pi(\Delta f/f_s)L}, \quad \widehat{\Delta f} = \frac{f_s}{2\pi L} \arg(P).$$

Unambiguous range $|\Delta f| < f_s/(2L)$ — smaller $L \rightarrow$ wider capture range but noisier. This is the same estimator OFDM uses (§5.6), where $L = N/2$.

Why residual CFO still matters. $\arg(\cdot)$ limits the estimate; whatever is left is a slow phase ramp the **phase tracker** (§5.5) or **OFDM pilot CPE** (§3.2) cleans up. Getting this balance wrong is exactly what smeared 16-QAM/OFDM over the air.

5.5 Carrier phase offset — estimation & PLL tracking

After timing + CFO the symbols still carry a residual phase $\varphi[n] = \varphi_0 + (\text{residual ramp})$. For **absolute** constellations (BPSK/QPSK/QAM) all of it must be removed before hard decisions; for **differential** schemes the static φ_0 cancels automatically (demod uses phase *differences*).

One-shot estimators (PhaseOffsetEstimator):

- **ML / preamble correlation** — optimal given the known preamble: $\hat{\varphi} = \arg(\sum_n r[n] p^*[n])$.
- **M-th power (blind, for M-PSK)** — raising to the M -th power collapses all M data phases onto one, removing the modulation: $\hat{\varphi} = \frac{1}{M} \arg(\sum_n r[n]^M)$.
- **Decision-directed** — $\hat{\varphi} = \arg(\sum_n r[n] \hat{s}^*[n])$ against the nearest constellation points $\hat{s}$ (used for QAM, after a coarse lock).

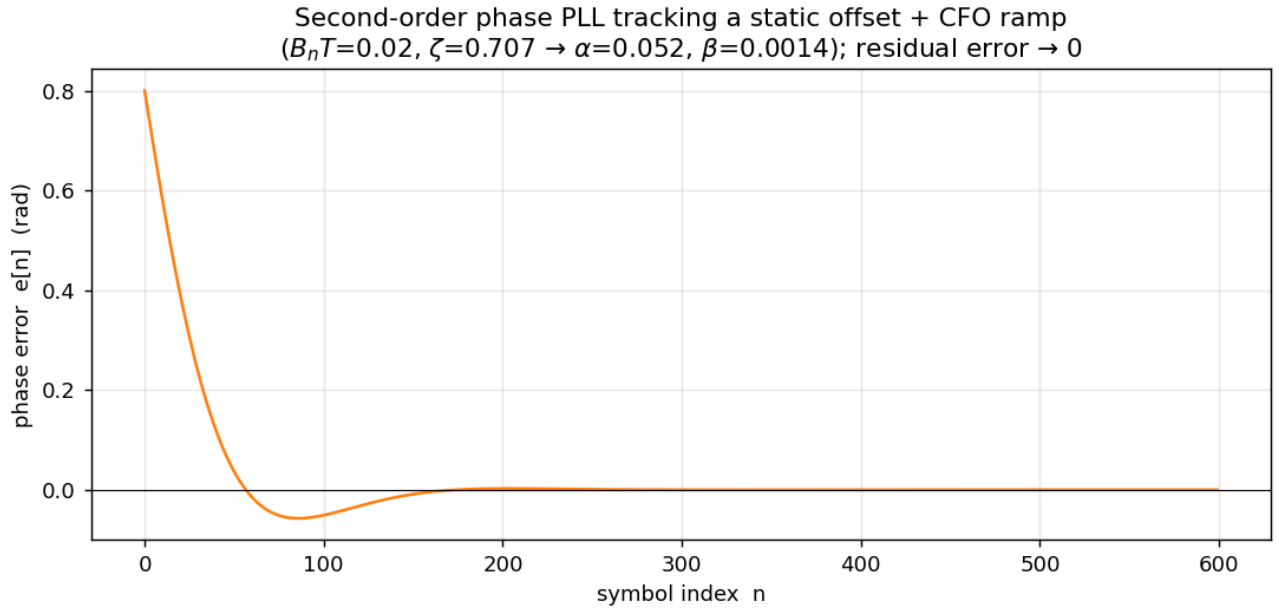
Tracking PLL (PhaseTracker) — a second-order digital PLL that follows a static offset **and** a constant frequency ramp (residual CFO). Per symbol: form the phase error against the nearest decision, then update an integrator (frequency) and the phase:

$$\begin{aligned} e[n] &= \arg(\text{corrected}[n] \hat{s}^*[n]) && \text{(phase detector, angle in } [-\pi, \pi]) \\ \text{freq} &+= \beta e[n] && \text{(integrator — tracks the residual CFO ramp)} \\ \varphi &+= \alpha e[n] + \text{freq} && \text{(NCO phase)} \end{aligned}$$

The loop coefficients come from the standard proportional-integral design for loop bandwidth $B_n T$ (--phase_loop_bw 0.02) and damping ζ (--phase_damping 0.707):

$$\theta_n = \frac{B_n T}{\zeta + 1/(4\zeta)}, \quad \alpha = \frac{4\zeta \theta_n}{1 + 2\zeta \theta_n + \theta_n^2}, \quad \beta = \frac{4\theta_n^2}{1 + 2\zeta \theta_n + \theta_n^2}.$$

($\beta = 0$ gives a first-order loop.) Smaller $B_n T$ = less jitter but slower pull-in; $\zeta = 0.707$ is the critically-damped standard.



Alternatives: Costas loop (joint carrier recovery on the raw samples), block Viterbi-Viterbi phase estimation, pilot-symbol-aided interpolation.

5.6 OFDM synchronization — Schmidl & Cox

OFDM recovers frame timing and CFO jointly from a sync symbol built as two identical time halves [A A] (generated by loading only the **even** subcarriers). Sliding a window of length $N/2$:

$$P(d) = \sum_{n=0}^{N/2-1} r[d+n] r^*[d+n+N/2] \quad \text{(half-to-half correlation)}$$

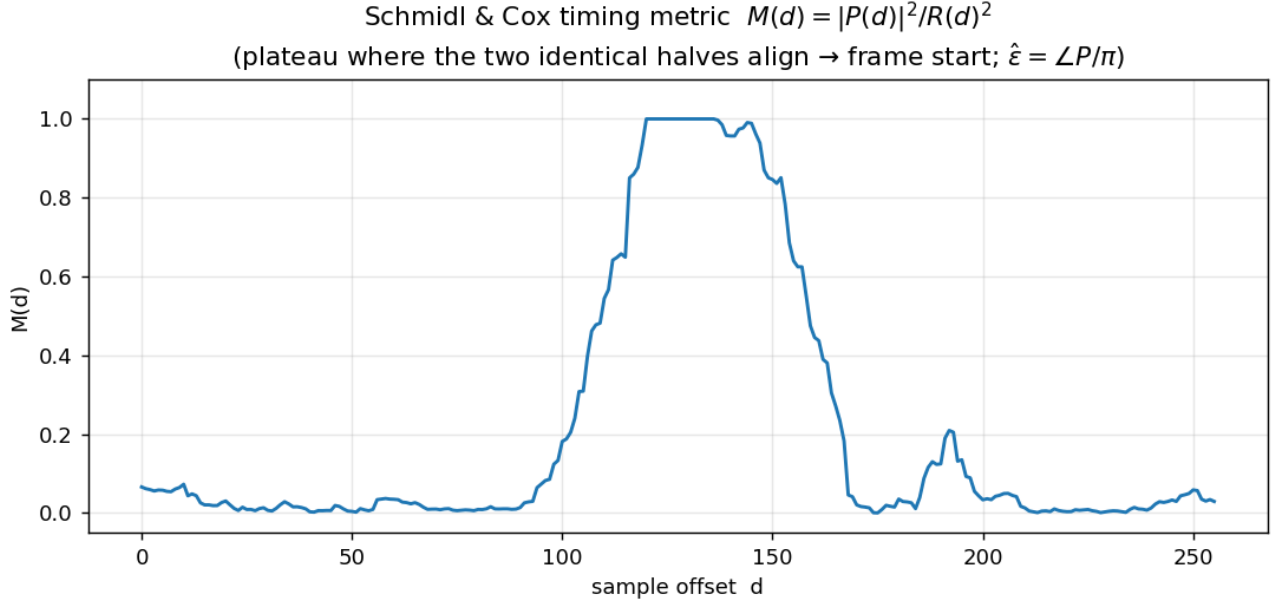
$$R(d) = \sum_{n=0}^{N/2-1} |r[d+n+N/2]|^2 \quad \text{(energy normaliser)}$$

$$M(d) = |P(d)|^2 / R(d)^2 \quad \text{(timing metric } \in [0, 1])$$

$M(d)$ forms a **plateau** where the two halves align; the frame start is the plateau's left edge plus a small guard into the CP (any offset inside the CP is just a per-subcarrier phase the channel estimate absorbs). The **fractional CFO** falls out of the same correlation:

$$\hat{\epsilon} = \arg(P) / \pi \quad (\text{subcarrier-spacing units, range } \pm 1 \text{ subcarrier}).$$

and the burst is derotated by $e^{-j2\pi\epsilon n/N}$. Residual CFO / phase drift across the frame is then tracked per symbol by the scattered-pilot CPE of §3.2. An energy gate ($R \geq 0.30 \max R$) stops the metric from locking onto the low-power noise pad before the burst.



Alternatives: Minn and Park metrics (sharper, no plateau ambiguity), CP-based blind sync.

6. Channel equalization (single-carrier)

--eq_type None (default) | LMS | RLS | DFE. --eq_taps 11, --eq_mu 0.3 (NLMS step), --eq_dd false (decision-directed tracking after training).

Math. A length- T linear FIR equalizer w estimates each symbol from a window of received samples, $\hat{s}[k] = w^H r[k] = \sum_i w_i^* r[k-i]$, choosing w to invert the channel h (ideally $w * h \approx \delta$). Two ways to find w :

- **Least-squares / MMSE training** (used, exact for the complex Zadoff-Chu preamble). Stack the preamble windows into a matrix R and the known symbols into s ; minimizing $\|Rw - s\|^2$ gives the normal equations

$$w = (R^H R + \lambda I)^{-1} R^H s$$

solved here by Gaussian elimination. λI = Tikhonov / diagonal-loading regularization: it bounds noise enhancement at channel nulls (the MMSE, not zero-forcing, solution).

- **NLMS adaptive update** (real preamble / decision-directed tracking): step down the instantaneous-error gradient, normalized by input power for stability:

$$e[k] = s[k] - \hat{s}[k], \quad w \leftarrow w + \mu \frac{e^*[k] r[k]}{\|r[k]\|^2 + \epsilon} \quad (\text{--eq_mu } 0.3, \ 0 < \mu < 2).$$

The LS solution places the main tap at the equalizer center, so a **center-tap group delay** of $(T - 1)/2$ samples must be compensated when aligning the output — mishandling this looked like "divergence" and was a fixed bug.

Status / why default None. On a clean cabled link no equalizer is needed. The LMS decision-directed loop currently **diverges on the real hardware signal** (DD error grows and destroys the symbols) — verified OTA where None decodes and LMS produces garbage. For frequency-selective channels, prefer **OFDM** (§3.2), which equalizes per-subcarrier without an adaptive FIR. Alternatives: DFE (feedback of past decisions, better on deep nulls), RLS (faster convergence than LMS, $O(n^2)$ cost), MLSE/Viterbi equalization (optimal, expensive).

7. Forward error correction (FEC)

`--fec true` (default off) — rate-1/2, constraint-length-7 **convolutional code** with **Viterbi** hard-decision decoding. Must match on both ends; halves the payload rate ($2\times$ symbols).

Math. Encoder: a 6-stage shift register holds the last $K - 1 = 6$ input bits (state $\sigma \in \{0, \dots, 63\}$); each input bit u emits two coded bits as mod-2 (XOR) taps of the register per the generator polynomials $G_1 = 171_8$, $G_2 = 133_8$ (the standard NASA / 802.11a code):

$$c_1 = G_1 \cdot [u, \sigma] \bmod 2, \quad c_2 = G_2 \cdot [u, \sigma] \bmod 2.$$

Each input bit thus influences $K = 7$ output-bit pairs (the constraint length / memory), which is what lets the decoder recover it even when some of those bits are flipped.

Viterbi decoding finds the maximum-likelihood transmitted sequence by the most-likely path through the 64-state trellis. For hard decisions the branch metric is the **Hamming distance** between the received pair and each branch's expected (c_1, c_2) ; the decoder keeps, per state, the survivor path with the smallest cumulative metric:

$$PM_t(\sigma') = \min_{\sigma \rightarrow \sigma'} \left[PM_{t-1}(\sigma) + d_H(r_t, c(\sigma \rightarrow \sigma')) \right]$$

then traces the survivors back to output the bits. It corrects any error pattern up to roughly $\lfloor (d_{\text{free}} - 1)/2 \rfloor$ — $d_{\text{free}} = 10$ for this code — per window; validated $\sim 100\%$ CRC-OK up to $\sim 2\%$ raw BER. (Two original bugs — encoder emitting from the *next* state, traceback storing the bit instead of the predecessor state — were fixed.)

Soft-decision Viterbi (Euclidean instead of Hamming metric, using the raw symbol distances) would add ~ 2 dB but isn't implemented.

Common alternatives: LDPC and Turbo codes (near-Shannon, used in Wi-Fi/5G), Reed-Solomon (burst errors), Polar codes (5G control), soft-decision Viterbi (~ 2 dB better than the hard-decision decoder here).

8. Error detection + ARQ

- **CRC-16-CCITT** appended to every chunk; the RX drops any chunk that fails the check (guarantees the delivered message is error-free).
- **Stop-and-wait ARQ** (`--role source_arq / sink_arq`): the source sends a chunk and waits for an ACK; on `--timeout 3000` ms it retransmits; it advances when ACKed and exits when all chunks are ACKed. The sink CRC-verifies each chunk, ACKs only verified ones, re-ACKs duplicates, and auto-stops once every chunk is in.
- **ACK transport** (`--ack-transport`):
 - `tcp` (default) — ACK over a socket; **no reverse RF path needed**. Source connects to `--ack-host 127.0.0.1` : `--ack-port 5599` (the sink listens there). Ideal when both radios share one host.
 - `rf` — ACK sent back over the air on the second RF path (needs a clean reverse link).

Math/theory. Treat the message bits as coefficients of a polynomial $M(x)$ over $\text{GF}(2)$; append 16 zeros ($M(x) \cdot x^{16}$) and divide by the CRC-16-CCITT generator $G(x) = x^{16} + x^{12} + x^5 + 1 \pmod{2}$ / XOR long division). The 16-bit remainder is the CRC; the RX recomputes it and flags any nonzero mismatch. Because an undetected error requires the error polynomial $E(x)$ to be an exact multiple of $G(x)$, this structure guarantees detection of: all single- and double-bit errors, any odd number of errors (G has the factor $x + 1$), and all burst errors of length ≤ 16 . Residual undetected probability $\approx 2^{-16}$ for random errors.

Stop-and-wait is the simplest ARQ (throughput limited by idling one round-trip per chunk). *Alternatives:* Go-Back-N / Selective-Repeat (pipelined, higher throughput), Hybrid-ARQ (FEC + ARQ combined — what you get here with `--fec on`: FEC corrects most errors, CRC+retransmit catches the rest).

One-way mode (`--role tx / rx`, no ACK): the TX cycles all chunks `--tx-reps` times; the RX reassembles from CRC-verified chunks and auto-stops `--rx-idle-timeout 8 s` after the last burst.

9. Automatic gain control (AGC)

`--AGC_type` Feed (feed-forward, default) or Closed (closed-loop). Feed-forward measures the detected burst's RMS and scales it to unit power in one shot (no loop lag); closed-loop adapts a gain over time. The energy detector's captured burst feeds the AGC before sync.

10. Visualization / EVM

`--viz true` (default) captures one TX and one RX burst and **auto-saves a figure on exit** to `<viz-dir>/<scheme>/figure.png` (per-modulation folder, `--viz-dir viz`). The 2×3 figure shows TX/RX **time domain**, **spectrum** (FFT), and **constellation** with the ideal points overlaid and an **EVM %** readout. Rough EVM ceilings for reliable decode: QPSK tolerates ~30–35% (more with FEC), 16-QAM needs <~12%, 64-QAM <~6%. Render manually with `python3 tools/plot_viz.py <dir> --fs 1.6e6 --save out.png`.

11. Complete command-line reference

Mode / roles

Option	Default	Controls
<code>--role</code>	both	tx, rx, both, source_arq, sink_arq
<code>--mode</code>	source	legacy source/sink selector
<code>--tx-reps</code>	20	one-way: times to cycle all chunks (no ACK)
<code>--rx-idle-timeout</code>	8.0	one-way RX: auto-stop after N s of no bursts (0 = until Ctrl-C)
<code>--num_bits</code>	1000	payload bits per packet
<code>--interval</code>	3000	TX gap between packets (ms)

ARQ / ACK

Option	Default	Controls
<code>--ack-transport</code>	tcp	ACK channel: tcp or rf
<code>--ack-host</code>	127.0.0.1	TCP: sink IP the source connects to
<code>--ack-port</code>	5599	TCP: ACK socket port
<code>--timeout</code>	3000	source ACK timeout (ms) before retransmit
<code>--timer_interval</code>	1000	sink ACK timer interval (ms)

Modulation / waveform / FEC

Option	Default	Controls
--scheme	QPSK	constellation (§2)
--fec	false	rate-1/2 K=7 conv + Viterbi (must match both ends)
--waveform	sc	sc (single-carrier) or ofdm
--ofdm-fft	64	OFDM subcarrier count (FFT size)
--ofdm-cp	16	OFDM cyclic-prefix length
--ofdm-tx-peak	0.5	OFDM peak scaling (PAPR / clipping guard)
--sps	2	samples/symbol (informational)

Preamble / pulse shaping

Option	Default	Controls
--preamble	m-sequence	m-sequence or zadoff
--m	5	m-sequence order (length $2^m - 1$)
--add_preamble	true	prepend preamble to each packet
--filter_type	rrc	rrc / rc / lp
--roll_off	0.25	RRC/RC roll-off β
--num_taps	151	filter tap count
--U / --D	2 / 1	pulse-shaper up/down sampling (wire sps = U/D)
--symbol_rate	0.8e6	symbol rate (Hz)
--num_threads	1	FFT threads

Energy detection

Option	Default	Controls
--alpha	0.95	IIR power-smoothing (larger = smoother)
--det-adaptive (--IIR_threshold_adaptive)	true	use noise_floor $\times$ mult vs fixed threshold
--det-mult (--IIR_threshold_multiplier)	5.0	adaptive threshold multiplier (raise OTA to 10–30)
--det-threshold (--energy_threshold)	1e-7	fixed threshold (adaptive off only)
--det-continuous	true	re-track noise floor during idle vs one-shot calib
--energy_packet_size	3300	samples to collect after detection (auto-sized)
--IIR_window_size	20	IIR window size

Synchronization / timing / phase

Option	Default	Controls
--sync-threshold (--sync_threshold)	15.0	ACQ correlation gate (set below true peak, above noise)
--sps_sync	5	samples/symbol at matched-filter output
--recv_msg_len	auto	data symbols to extract per burst (auto from scheme)
--timing_loop_bw	0.015	Gardner TED loop bandwidth BnT
--timing_damping	0.707	Gardner TED damping
--phase_loop_bw	0.02	carrier phase PLL bandwidth
--phase_damping	0.707	carrier phase PLL damping

Equalizer

Option	Default	Controls
--eq_type	None	None / LMS / RLS / DFE (see §6 — LMS diverges OTA)
--eq_taps	11	equalizer tap count
--eq_mu	0.3	NLMS step (DD / real-preamble training)
--eq_dd	false	decision-directed tracking after LS training

RF hardware (TX)

Option	Default	Controls
--tx-args	""	UHD TX device (e.g. serial=30CD424)
--tx-rate	1.6e6	TX sample rate (Hz) = symbol_rate·U/D
--tx-freq	2.412e9	TX centre frequency (Hz)
--tx-gain	20	TX gain (dB)
--tx-bw	1.0e6	TX analog bandwidth (Hz)
--tx-ant	TX/RX	TX antenna port
--tx-subdev	A:A	TX subdev spec
--tx-channel	0	TX channel index

RF hardware (RX)

Option	Default	Controls
--rx-args	""	UHD RX device (e.g. serial=30CD3F7)
--rx-rate	1.6e6	RX sample rate (Hz), must equal TX rate
--rx-freq	2.412e9	RX centre frequency (Hz)
--rx-gain	30	RX gain (dB) — key for EVM/clipping (§ recipes)
--rx-bw	1.0e6	RX analog bandwidth (Hz)
--rx-ant	RX2	RX antenna port
--rx-subdev	A:A	RX subdev spec
--rx-channel	0	RX channel index

Common RF / buffering / misc

Option	Default	Controls
--ref	internal	clock reference: internal / external / mimo
--settling	0.2	RF settling time (s)
--uhd_timeout	1000	UHD TX timeout (ms)
--samps_per_buff	10000	samples per UHD receive buffer
--num_recv_request	0	total samples to receive (0 = continuous)
--AGC_type	Feed	Feed (feed-forward) or Closed (closed-loop)
--skip-rate-check	off	bypass startup rate-consistency check
--viz	true	capture + auto-save TX/RX figure on exit
--viz-dir	viz	base dir for figures (per-scheme subfolder made)

12. Verified command recipes (915 MHz, VERT900 antennas)

QPSK OFDM + ARQ (reliable OTA demo). Start the sink first.

```
# Sink / RX
./sdr_system --role sink_arq --rx-args serial=30CD3F7 --tx-args serial=30CD3F7 \
--rx-subdev A:A --rx-ant RX2 --rx-freq 915e6 --rx-rate 1.6e6 --rx-gain 25 \
--waveform ofdm --ofdm-fft 64 --ofdm-cp 16 --scheme QPSK --fec true \
--ack-transport tcp --ack-port 5599 --det-mult 3 --viz-dir viz
# Source / TX
./sdr_system --role source_arq --tx-args serial=30CD424 --rx-args serial=30CD424 \
--tx-subdev A:A --tx-ant TX/RX --tx-freq 915e6 --tx-rate 1.6e6 --tx-gain 82 \
--waveform ofdm --ofdm-fft 64 --ofdm-cp 16 --ofdm-tx-peak 0.5 --scheme QPSK --fec true \
--ack-transport tcp --ack-host 127.0.0.1 --ack-port 5599 --timeout 3000 --viz-dir viz
```

Single-carrier QPSK one-way (swap `--waveform ofdm ...` for `--waveform sc`, roles rx/tx, add `--tx-reps 20`).

16-QAM / 64-QAM: same as above with `--scheme 16QAM` (or `64QAM`) on both ends — but needs much higher SNR (EVM < ~12% / 6%). Use a cabled link (SMA + 30–40 dB attenuator) or very close antennas; tune `--rx-gain` so the RX Peak= stays under ~0.8 (below 1.0 = ADC clipping) while RMS is as high as possible. Best OTA point found so far: `--rx-gain 21 --tx-gain 80 --ofdm-tx-peak 0.45`.

Level-tuning cheatsheet (the #1 cause of "nothing decodes")

- RX log Peak > 1.0 -> **clipping**, lower `--rx-gain` (fatal for QAM).
- No detections (bursts=0) -> signal too weak, raise `--tx-gain` or `--rx-gain`, or lower `--det-mult`.
- Detections but CRC fails -> check the constellation EVM in the figure; if the order is too high for the link SNR, drop to QPSK or improve the link.
- Keep `--fec true` on marginal links; it closes the last few % of bit errors.